

Shuvayu Roy

CONTACT INFORMATION	Department of Physics, Indian Institute of Technology, Gandhinagar, Gujarat 382355, India <i>Email:</i> shuvayu.roy@iitgn.ac.in shuvayu.roy.physics@gmail.com InspireHEP: inspirehep.net/authors/2140814 ORCID: 0000-0002-5725-5712
DATE OF BIRTH	October 5, 1996
NATIONALITY	Indian
RESEARCH INTERESTS	Relativistic hydrodynamics, Black hole Thermodynamics, Fluid-Gravity Correspondence, Holography, Higher-Derivative Gravity Theories, Effective Field Theories.
EMPLOYMENT HISTORY	Indian Institute of Technology, Gandhinagar • Position: Postdoctoral Fellow • Dates: September 30, 2024 - September 29, 2026 • Type of Appointment: Fixed-term, Full-time
EDUCATION	Integrated M.Sc.-Ph.D.: Homi Bhabha National Institute • Constituent Institution: National Institute of Science Education and Research (NISER), Bhubaneswar, Odisha, India. PIN - 752050, India • Discipline: Physics • Thesis Advisor: Dr. Sayantani Bhattacharyya • Title of the Thesis: “<i>Black Holes and Relativistic Fluids from the perspective of Near-Equilibrium Dynamics</i>” • Date of Awarding of Degree: December 3, 2024. Bachelor of Science (B.Sc.): University of Calcutta • College: Scottish Church College, 1 & 3, Urquhart Square, Manicktala, Kolkata, West Bengal, India. PIN - 700006 • Honours: Physics, General: Mathematics and Chemistry • Passed: 2017
PUBLICATIONS	1. S. Roy, S. Mitra, R. Singh, “<i>Necessary conditions for causality from linearized stability at ultra-high boosts</i>”, [arXiv:2605.12291] 2. S. Acharya, S. Roy, S. Sarkar, “<i>Stochastic Evolution of Primordial Black Holes to near-extremality in EFTs of Gravity</i>”, <i>Phys.Rev.D</i> 114 (2026) 2, 2, [arXiv:2602.21923] 3. S. Bhattacharyya, S. Mitra, S. Roy, R. Singh, “<i>Relativistic Dispersion Spectra across Lorentz boosted frames: Spurious modes and the enigma of causality</i>”, <i>JHEP</i> 05 (2026) 215, [arXiv:2512.12450] 4. C. Sharma, S. Roy, S. Sarkar, “<i>Ladder Symmetry: The Necessary and Sufficient Condition for Vanishing Love Numbers</i>”, <i>Phys.Rev.D</i> 113 (2026) 2, 2 [arXiv:2511.09670] 5. S. Acharya, S. Roy, S. Sarkar, “<i>EFT Corrections to Majumdar-Papapetrou Black Holes</i>”, <i>Phys.Rev.D</i> 112 (2025) 8, 084043, [arXiv:2503.19646]

6. S. Bhattacharyya, S. Mitra, **S. Roy**, R. Singh, “*Relativistic fluid dynamics in a ‘hydro’ frame*”, [arXiv:2410.19015]
7. S. Bhattacharyya, S. Mitra, **S. Roy**, R. Singh, “*Field redefinition and its impact in relativistic hydrodynamics*”, Phys.Rev.D 111 (2025) 1, 014034, [arXiv:2409.15387]
8. S. Bhattacharyya, S. Mitra, **S. Roy**, “*Causality and stability in relativistic hydrodynamic theory - a choice to be endured*”, Phys.Lett.B 856 (2024) 138918, [arXiv:2407.18997]
9. A. Kar, P. Dhivakar, **S. Roy**, B. Panda, A. Shaikh “*Iyer-Wald ambiguities and gauge covariance of Entropy current in Higher derivative theories of gravity*”, JHEP07(2024)016, [arXiv:2403.04749]
10. S. Bhattacharyya, S. Mitra, **S. Roy** “*Frame transformation and first order stable-causal hydrodynamic theory*”. [arXiv:2312.16407]
11. **S. Roy**, S. Mitra, “*Causality Criteria from Stability Analysis at Ultra-High Boost*”. [arXiv:2306.07564] Phys.Rev.D 110 (2024) 1, 016012
12. Chandranathan A, S. Bhattacharyya, M. Patra, **S. Roy**, “*Entropy Current and Fluid-Gravity Duality in Gauss-Bonnet theory*”, JHEP09(2023)070, [arXiv:2208.07856]
13. S. Bhattacharyya, P. Jethwani, M. Patra, **S. Roy**, “*Reparametrization Symmetry of Local Entropy Production on a Dynamical Horizon*”, Phys.Rev.D 108 (2023) 10, 104032, [arXiv:2204.08447]
14. S. Bhattacharyya, P. Dhivakar, A. Dinda, N. Kundu, M. Patra, **S. Roy**, “*An entropy current and the second law in higher derivative theories of gravity*”, JHEP09(2021)169, [arXiv:2105.06455]

COMPUTATIONAL SKILLS

1. Mathematica (including the xAct suite of packages)
2. Python

TEACHING EXPERIENCE (TEACHING ASSISTANT)

Black Hole Mechanics and Thermodynamics in Effective Field Theories of Gravity (2026)

Instructor: Dr. Nilay Kundu

IAGRG School on Gravitation and Cosmology, IUCAA

Wald’s Covariant Phase Space Formalism (2023)

Instructor: Dr. Sudipta Sarkar

IAGRG School on Gravitation and Cosmology, ICTS-TIFR

P304: Special Theory of Relativity (2022)

Instructor: Dr. Sayantani Bhattacharyya

NISER

P743 + P341: Nuclear Physics and Instrumentation Lab (2022)

Instructors: Dr. Prolay Kumar Mal and Dr. Pratap Kumar Sahoo

NISER

P453: Quantum Field Theory I (2021)

Instructor: Dr. Sayantani Bhattacharyya

NISER

P206: Quantum Mechanics I (2021)

Instructor: Dr. Nishikanta Khandai

NISER

SOME OF THE
WORKSHOPS AND
CONFERENCES
ATTENDED

- Frontiers in Gravity 2026, *IISER Mohali, (March 2026)*, [**Oral Presentation**]
- Hydrodynamics, Fluctuations, and Noise In quantum and classical systems, *ICTS-TIFR, December, 2025*, [**Oral Presentation**]
- Theoretical Tools for Gravitational Wave Physics, *ETH Zurich, (November, 2025)*, [**Flash Talk, Poster**]
- Monsoon Meet on Gauge-Gravity, *IISc., (August, 2025)*, [**Oral Presentation**]
- Field Theory and Turbulence, *ICTS-TIFR (December 2023)*, [**Poster**]
- Indian Strings Meeting 2023, *IIT Bombay (December 2023)*, [**Oral Presentation**]
- New Perspectives in Conformal Field Theory and Gravity, *DESY Hamburg (September 2023)*, [**Oral Presentation**]
- Hydrodynamics at All Scales, *NORDITA (September 2023)*
- Students Talks in Trending Topics in Theory (ST⁴ 2023), *IIT Mandi (July 2023)*, [**Member of Organising Committee**]
- Kavli Asian Winter School, *IBS, Daejeon, Korea (January, 2023)*, [**Gong Show, Poster**]

ACADEMIC
ACHIEVEMENTS

Debiprasad Ray Chaudhuri Memorial Prize, 2015

Dept. of Physics, Scottish Church College, for Best Seminar Presentation.

Anil Sen Memorial Prize, 2017

Dept. of Physics, Scottish Church College, for securing the highest marks in practical examination (Part I, Part II, Part III taken together) conducted by the University of Calcutta

PERSONAL
INTERESTS

Playing Musical Instruments, Photography, Birdwatching

REFERENCES

Dr. Sudipta Sarkar

Department of Physics, IIT Gandhinagar, India

Email: sudiptas@iitgn.ac.in

Dr. Sayantani Bhattacharyya

School of Mathematics, University of Edinburgh, United Kingdom

Email: sbhatta5@ed.ac.uk

Dr. Nilay Kundu

Department of Physics, IIT Kanpur, India

Email: nilayhep@iitk.ac.in

Dr. Sukanya Mitra

School of Physical Sciences, NISER, Bhubaneswar, India

Email: sukanya.mitra10@gmail.com